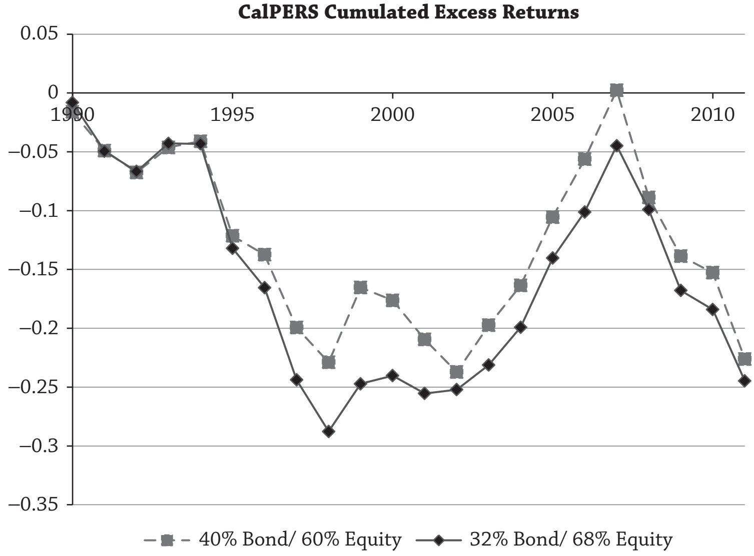

Figure 10.3

Figure 10.3 plots cumulative excess returns for CalPERS. The estimated 32% stocks/68% bonds benchmark portfolio is shown in the solid line and a standard 40% stocks/60% bonds portfolio is overlaid for comparison. Note the similarity. CalPERS performance improves during 2000–2007. But during the financial crisis in 2008, things completely fall apart, and the fund's performance continued to deteriorate in 2010 and 2011. A large part of this dismal showing was due to CalPERS' failure to rebalance in 2008 and 2009: it sold equities rather than buying them when prices were low (see chapter 4 on the optimality of rebalancing as a long-term strategy).

Real Estate

Canada Pension Plan considers real estate to have many characteristics in common with fixed income and equities—so much in common that the plan doesn't consider real estate a separate asset class. But can real estate exposure be replicated by a factor portfolio of stocks, bonds, and, potentially, listed REITs (see chapter 11), which offer indirect real estate exposure?

Real estate returns are complicated because they are not tradeable. Leaving aside this problem—which I deal with in chapter 11 along with other illiquid investments—I take quarterly real estate returns from the National Council of Real Estate Investment Fiduciaries from June 1978 to December 2011 (my left-hand side variable). I consider factor benchmark regressions using S&P 500 stock returns, Ibbotson long-term corporate bond returns, and the FTSE NAREIT index returns (my right-hand side variables).